Notation

In the following note, I will denote the basis of $M_m(\mathbb{C})$ by $\{E_{ij}\}_{i,j=1}^{m}$ and the basis of $M_n(\mathbb{C})$ by $\{F_{\alpha\beta}\}_{\alpha\beta=1}^{n}$. Next, we intoduce some notation that will be constantly appearing in the text.

Let $\varphi : M_m \to M_n$ be some Hermitian preserving maps. Then, there according to choi jamiolkowski isomorphism, we have its corresponding so called choi matrix defined as

$$C_\varphi := \sum_{i,j=1}^{m} \varphi(E_{ij}) \otimes E_{ij} \in M_n \otimes M_m$$

Note that for Hermitian preserving maps, its choi mathrix C_φ is Hermitian.

Since C_φ is a Hermitian, it follows that there exist an eigen decomposition of C_φ

$$C_\varphi = \sum_{i=1}^{mn} \lambda_i v_i v_i^* = \sum_{t=1}^{r} \rho_t \sum_{j\in J_t} v_j v_j^* = \sum_{t=1}^{r} \rho_t P_t$$

Next, we define the following notation:

$$Q_t := (\mathrm{id} \otimes \mathrm{tr})(P_t)$$
$$W_i W_j^* := (\mathrm{id} \otimes \mathrm{tr})(v_i v_j^*)$$

Last, it is important to keep in mind that for Upper letter trace Tr, it means the normal trace operation and the lower letter trace tr means the normalized trace operation, i.e. in the case of underlying space is M_n, we have $\mathrm{tr}_n = \frac{1}{n}\mathrm{Tr}$.

Also, we will use the symbol $(\cdot,\cdot)$ as inner product with the complex conjugate on the second slot.

Some useful calculation

In this section, I will provide some useful lemma to help us do the calculation.

Lemma 1. For all $1 \le i, j \le mn$, we have

$$\mathrm{Tr}(W_i W_j^*) = \delta_{ij}$$

Proof. This can be seen as $\mathrm{Tr}\big((\mathrm{id} \otimes \mathrm{tr})(v_i v_j^*)\big)$. Since the trace operation is linear and the fact that $\{v_i\}_{i=1}^{mn}$ is the orthonormal basis of $M_n \otimes M_m$, we get $\mathrm{Tr}\big((\mathrm{id} \otimes \mathrm{tr})(v_i v_j^*)\big) = \mathrm{Tr}(v_i v_j^*) = (v_i, v_j) = \delta_{ij}$ □

Lemma 2. For all $1 \le i, j \le mn$, we have

$$Q_t = \sum_{j\in J_t} W_j W_j^*$$

Proof. This can be seen directly from the definition. □

Lemma 3. We will denote the multiplicity of eigenvalue ρ_t as r_t. We should observe that $r_t = |J_t| = rank(P_t) = \mathrm{Tr}(Q_t)$

Proof. We shall show the last equality. This follows by the observation $\mathrm{Tr}(Q_t) = \mathrm{Tr}((\mathrm{id} \otimes \mathrm{tr})(P_t)) = \mathrm{Tr}(P_t) = rank(P_t)$ □

Main calculation

Our goal is to give a closed form of the equation

$$\lim_{d\to\infty} \mathrm{tr}_{nd}\Big(\big(X_d^{\varphi}\big)^l\Big) = \frac{m}{n} \sum_{k_1,k_2,\dots k_l=1}^{mn} \lambda_{k_1}\lambda_{k_2}\dots\lambda_{k_l} \sum_{\pi\in NC(l)} \kappa_\pi[x]\,\mathrm{tr}_{K(\pi)}\big[A_{k_1},\dots A_{k_l}\big]$$

Theorem 4. For any $\pi \in NC(l)$ and $Q_t = \frac{r_t}{n} I = d_t I$ for all t, we have

$$\frac{m}{n} \sum_{k_1,k_2,\dots k_l=1}^{mn} \lambda_{k_1}\dots\lambda_{k_l}\,\mathrm{tr}_{K(\pi)}\big[A_{k_1},\dots A_{k_l}\big] = \prod_{V\in\pi}\left(\sum_{t=1}^{r} \frac{d_t \rho_t^{|V|}}{m^{|V|-1}}\right)$$

Proof. We will start by interchanging the summation order by

$$\sum_{k_1,k_2,\dots k_l=1}^{mn} \lambda_{k_1}\dots = \sum_{i=1}^{l}\sum_{t_i=1}^{r}\sum_{k_i\in J_{t_i}} \lambda_{k_1}\dots = \sum_{V\in\pi}\sum_{i\in V}\sum_{t_i=1}^{r} \rho_{t_1}\dots \sum_{k_i\in J_{t_i}}.$$

Under this reorder, we will focus on different cases when taking specific block V of a partition.

There are only two cases of the blocks. Since the partition is a non-crossing partition, we can guarantee that in each step. The block V we can choose has the interval or singleton property. Lets consider the first case.

Suppose $V = (u, u+1, \dots u+h)$, by calculating its complement of the index $u, u+1, \dots u+h$, we can acquire it is of the form $\mathrm{tr}_m\big(A_{k_u}\big)\,\mathrm{tr}_m\big(A_{k_{u+1}}\big)\dots$ $\mathrm{tr}_m\big(A_{k_{u+h-1}}\big)$. Writing out all the trace blocks as $\mathrm{tr}_m\big(W_{k_u}^* W_{k_{u+1}}\big)$ and apply Lemma 2, we will get $k_u = k_{u+1} = \dots = k_{u+h}$ with each trace block also contributing to $\frac{1}{m}$, since it is in the normalized trace operation.

Note that also we have the trace block containing k_{u-1} and k_{u+h}. In this trace block, it is of the form

$$\mathrm{tr}_m\big(B_1 A_{k_{u-1}} A_{k_{u+h}} B_2\big) = \mathrm{tr}_m\big(B_1 W_{k_{u-1}}^* W_{k_u} W_{k_{u+h}} W_{k_{u+h+1}} B_2\big)$$

Since $k_u = k_{u+h} \in J_{t_u}$, it can be decompose into $\mathrm{tr}\big(B_1 W_{k_{u-1}}^* Q_{t_u} W_{k_{u+h+1}} B_2\big) = d_{t_u}\,\mathrm{tr}\big(B_1 W_{k_{u-1}}^* W_{k_{u+h+1}} B_2\big)$. Putting these in together, given $V \in \pi$, we obtain

$$\sum_{i\in V}\sum_{t_i=1}^{r}\rho_{t_1}...\rho(t_r)\sum_{k_i\in J_{t_i}}\frac{d_{t_u}}{m^{|V|-1}}\delta_{k_uk_{u+1}}...\delta_{k_{u+h-1}k_{u+h}}\operatorname{tr}_{K(\pi)}\left[A_{k_1},..\widehat{A_{k_u}}...\widehat{A_{k_{u+h}}}...A_{k_l}\right]$$
$$=\left(\sum_{t=1}^{r}\frac{d_t\rho_t^{|V|}}{m^{|V|-1}}\right)\sum_{t_i=1}^{r}\rho_{t_1}..\widehat{\rho_{t_u}}...\widehat{\rho_{t_{u+h}}}..\rho_{t_l}\operatorname{tr}_{K(\pi)}\left[A_{k_1},..\widehat{A_{k_u}}...\widehat{A_{k_{u+h}}}...A_{k_l}\right]$$

Similarly, we do the same reduction to the singleton block, say $V=(u)$. Its complement index will have A_{u-1} and A_u being in the same trace block. That is of the form $\operatorname{tr}_m(B_1W_{u-1}^*W_uW_u^*W_{u+1}B_2)$. In this case, summing over $\{u\}$ will get $d_{t_u}\operatorname{tr}_m(B_1W_{u-1}^*W_{u+1}B_2)$. Note that we have $1=\frac{1}{m^{|V|-1}}$. Thus, we still contribute of the same form.

$$\sum_{t=1}^{r}\frac{d_t\rho_t^1}{m^0}\rho_{t_1}...\widehat{\rho_{t_u}}...\rho_{t_l}\operatorname{tr}_{K(\pi)}\left[A_{k_1},...\widehat{A_{k_u}},...A_{k_l}\right]$$

Inductively doing the same procedure will arrive at the final step, which only remains one-block case.

In this case, we have the complement of all one-size trace block. That is,

$$\operatorname{tr}_{K(\pi)}\left[A_{k_1},A_{k_2},...A_{k_l}\right]=\operatorname{tr}\left(A_{k_1}\right)...\operatorname{tr}\left(A_{k_l}\right)=\frac{1}{m^l}\delta_{k_1k_2}...\delta_{k_lk_1}$$

Then summing together gets

$$\frac{\rho^{|\pi|}|J_\pi|}{m^l}=\frac{\rho^{|\pi|}nd_t}{m^l}$$

Adding with $\frac{n}{m}$, we get the form of

$$\frac{\rho^{|\pi|}d_t}{m^{l-1}}$$

Combining all procedures, we arrive at

$$\frac{m}{n}\sum_{k_1,k_2,...k_l=1}^{mn}\lambda_{k_1}...\lambda_{k_l}\operatorname{tr}_{K(\pi)}\left[A_{k_1},...A_{k_l}\right]=\prod_{V\in\pi}\left(\sum_{t=1}^{r}\frac{d_t\rho_t^{|V|}}{m^{|V|-1}}\right)$$

□

Note that under the assumption of $Q_t=d_tI$ for all $t\in[r]$, we will obtain the closed form of the main equation.

$$\mu^\varphi=\boxplus_{t=1}^{r}\left(D_{\frac{\rho_t}{m}}\mu\right)^{\boxplus md_t}$$

or equivalently,

$$\kappa_l(x^{\varphi}) = \sum_{t=1}^{r} m d_t \left(\frac{\rho_t}{m}\right)^l \kappa_l(\mu)$$

Some computation results

We will evaluate for some moments of the lower dimension, especially for $l = 3$.

In this case, there are five non crossing partitions. We will compute the exact form of

$$\frac{m}{n} \sum_{k_1,k_2,\ldots k_l=1}^{mn} \lambda_{k_1}\lambda_{k_2}\ldots\lambda_{k_l} \sum_{\pi\in NC(l)} \kappa_\pi[x] \operatorname{tr}_{K(\pi)}\left[A_{k_1},\ldots A_{k_l}\right]$$

For $\pi = (1,2,3)$, we have $K(\pi) = (1)(2)(3)$. Then, we will derive

$$\begin{aligned}
&\kappa_3(x) \sum_{k_1,k_2,k_3=1}^{mn} \lambda_{k_1}\lambda_{k_2}\lambda_{k_3} \operatorname{tr}_m(A_1)\operatorname{tr}_m(A_2)\operatorname{tr}_m(A_3) \\
&= \frac{\kappa_3(x)}{m^2} \sum_{k_1,k_2,k_3=1}^{mn} \lambda_{k_1}\lambda_{k_2}\lambda_{k_3} \operatorname{tr}_m(A_1)\delta_{k_2k_3}\delta_{k_3k_1} \\
&= \frac{n}{m}\frac{\kappa_3(x)}{m^2} \sum_{k_1=1}^{mn} \lambda_{k_1}^3 \operatorname{tr}_n\left(W_{k_1}W_{k_1}^*\right) = \frac{n\kappa_3(x)}{m^3} \operatorname{tr}_n\left((\mathrm{id}\otimes\operatorname{tr}_m)\left(C_\varphi^3\right)\right)
\end{aligned}$$

For $\pi = (1,2)(3)$, we have $K(\pi) = (1)(2,3)$. We have

$$\begin{aligned}
&\kappa_2(x)\kappa_1(x) \sum_{k_1,k_2,k_3=1}^{mn} \lambda_{k_1}\lambda_{k_2}\lambda_{k_3} \operatorname{tr}_m\left(A_{k_1}\right)\operatorname{tr}_m\left(A_{k_2}A_{k_3}\right) \\
&= \kappa_2(x)\kappa_1(x) \sum_{k_1,k_2,k_3=1}^{mn} \lambda_{k_1}\lambda_{k_2}\lambda_{k_3}\delta_{k_1k_2} \operatorname{tr}_m\left(A_{k_2}A_{k_3}\right) \\
&= \frac{\kappa_2(x)\kappa_1(x)}{m} \sum_{k_1,k_3=1}^{mn} \lambda_{k_1}^2\lambda_{k_3} \operatorname{tr}_m\left(W_{k_1}^*W_{k_3}W_{k_3}^*W_{k_1}\right) \\
&= \frac{n\kappa_2(x)\kappa_1(x)}{m^2} \operatorname{tr}_n\left((\mathrm{id}\otimes\operatorname{tr}_m)\left(C_\varphi\right)(\mathrm{id}\otimes\operatorname{tr}_m)\left(C_\varphi^2\right)\right)
\end{aligned}$$

Since the index has some symmetry in the other two cases, that is when $\pi = (1)(2,3), (1,3)(2)$, the computed results will multiply by 3.

Now for the last case. $\pi = (1)(2)(3)$ and $K(\pi) = (1,2,3)$. We have

$$\kappa_1(x)^3 \sum_{k_1,k_2,k_3=1}^{mn} \lambda_{k_1}\lambda_{k_2}\lambda_{k_3} \operatorname{tr}_m\Big(A_{k_1}A_{k_2}A_{k_3}\Big)$$
$$= \kappa_1(x)^3 \sum_{k_1,k_2,k_3=1}^{mn} \lambda_{k_1}\lambda_{k_2}\lambda_{k_3} \operatorname{tr}_m\Big(W_{k_1}^* W_{k_2} W_{k_2}^* W_{k_3} W_{k_3}^* W_{k_1}\Big)$$
$$= \frac{n\kappa_1(x)^3}{m} \operatorname{tr}_n\Big((\mathrm{id}\otimes \operatorname{tr}_m)\big(C_\varphi\big)^3\Big)$$

Combining all the cases and denote $\mathbb{E}(X) = (\mathrm{id}\otimes \operatorname{tr}_m)(X)$ for short, and the results will be

$$\operatorname{tr}_n\Big(\frac{n\kappa_1(x)^3}{m}\mathbb{E}\big(C_\varphi\big)^3 + \frac{3n\kappa_2(x)\kappa_1(x)}{m^2}\mathbb{E}\big(C_\varphi^2\big)\mathbb{E}\big(C_\varphi\big) + \frac{n\kappa_3(x)}{m^3}\mathbb{E}\big(C_\varphi^3\big)\Big) =$$
$$\operatorname{Tr}_n\Big(\frac{\kappa_1(x)^3}{m}\mathbb{E}\big(C_\varphi\big)^3 + \frac{3\kappa_2(x)\kappa_1(x)}{m^2}\mathbb{E}\big(C_\varphi^2\big)\mathbb{E}\big(C_\varphi\big) + \frac{\kappa_3(x)}{m^3}\mathbb{E}\big(C_\varphi^3\big)\Big)$$

To group it properly, we see that it is the same as

$$\operatorname{Tr}_n\Big(\Big(\frac{1}{m}\mathbb{E}\big(\kappa_1(x)C_\varphi\big)^3 + \frac{3}{m^2}\mathbb{E}\big(\kappa_2(x)C_\varphi^2\big)\mathbb{E}\big(\kappa_1(x)C_\varphi\big) + \frac{1}{m^3}\mathbb{E}\big(\kappa_3(x)C_\varphi^3\big)\Big)$$

Next, we look at the case where $l = 4$. In this situation, there are 14 non-crossing partitions. Thus, we will eliminate the unnecessary computation and only give the important calculation.

Most of the partitions share the same partitions. We will choose one in each partition as representative.

For $\pi = (1)(2)(3)(4)$, $K(\pi) = (1,2,3,4)$. In this case, one could easily calculate the corresponding equation be

$$\kappa_1^4 \frac{n}{m} \operatorname{tr}_n\Big((\mathrm{id}\otimes \operatorname{tr}_m)\big(C_\varphi\big)^4\Big)$$

For $\pi = (1,2)(3)(4)$ and $K(\pi) = (1)(1,2,3)$. We have

$$\kappa_2\kappa_1^2 \frac{n}{m^2} \operatorname{tr}_n\Big((\mathrm{id}\otimes \operatorname{tr})\big(C_\varphi^2\big)(\mathrm{id}\otimes \operatorname{tr})\big(C_\varphi\big)^2\Big)$$

In this case, the computations are the same for $\pi = (1)(2,3)(4)$, $\pi = (1)(2)(3,4)$, $\pi = (1,4)(2)(3)$.

For $\pi = (1,3)(2)(4)$ and $K(\pi) = (1,2)(3,4)$. We can get

$$\kappa_2\kappa_1^2 \sum_{k_1,k_3}^{mn} \lambda_{k_1}\lambda_{k_3} \frac{n^2}{m^2} \operatorname{tr}_n\Big((\mathrm{id}\otimes \operatorname{tr}_m)\big(C_\varphi\big) W_{k_3}^* W_{k_1}\Big) \operatorname{tr}_n\Big((\mathrm{id}\otimes \operatorname{tr}_m)\big(C_\varphi\big) W_{k_1}^* W_{k_3}\Big)$$

This case is the same as $\pi = (1)(2,4)(3)$

For $\pi = (1,2,3)(4)$ and $K(\pi) = (1)(2)(3,4)$. We have

$$\kappa_3\kappa_1\frac{n}{m^3}\operatorname{tr}_n\Big((\mathrm{id}\otimes\operatorname{tr}_m)\big(C_\varphi^3\big)(\mathrm{id}\otimes\operatorname{tr}_m)\big(C_\varphi\big)\Big)$$

In this case, the computations are the same for $\pi = (1,2,4)(3)$, $\pi = (1,3,4)(2)$, $\pi = (1)(2,3,4)$.

For $\pi = (1,4)(2,3)$ and $K(\pi) = (1,3)(2)(4)$. We have

$$\kappa_2^2\frac{n}{m^3}\operatorname{tr}_n\Big((\mathrm{id}\otimes\operatorname{tr}_m)\big(C_\varphi^2\big)^2\Big)$$

This is the same for $\pi = (1,2)(3,4)$.

The last case is $\pi = (1,2,3,4)$ with $K(\pi) = (1)(2)(3)(4)$. In this situation, we will get

$$\kappa_4\frac{n}{m^4}\operatorname{tr}_n\Big((\mathrm{id}\otimes\operatorname{tr}_m)\big(C_\varphi^4\big)\Big)$$

By grouping together and replacing $\mathrm{id}\otimes\operatorname{tr}_m = \mathbb{E}$, we get,

$$\operatorname{Tr}_n\Big(\frac{1}{m}\mathbb{E}\big(\kappa_1C_\varphi\big)^4 + \frac{4}{m^2}\mathbb{E}\big(\kappa_2C_\varphi^2\big)\mathbb{E}\big(\kappa_1C_\varphi\big)^2 +$$
$$\frac{4}{m^3}\mathbb{E}\big(\kappa_3C_\varphi^3\big)\mathbb{E}\big(\kappa_1C_\varphi\big) + \frac{2}{m^3}\mathbb{E}\big(\kappa_2C_\varphi^2\big)^2 + \frac{1}{m^4}\mathbb{E}\big(\kappa_4C_\varphi^4\big)\Big) +$$
$$2\sum_{k_1,k_3=1}^{mn}\lambda_{k_1}\lambda_{k_3}\kappa_2\frac{n^2}{m^2}\operatorname{tr}_n\Big(\mathbb{E}\big(\kappa_1C_\varphi\big)W_{k_3}^*W_{k_1}\Big)\operatorname{tr}_n\Big(\mathbb{E}\big(\kappa_1C_\varphi\big)W_{k_1}^*W_{k_3}\Big)$$

Fock space

To extend our discussions, we will introduce the so called Fock space.

Definition 1. Given a Hilbert space $\mathcal{H}$. We define the Fock space by its direct sum of tensor product. For simplicity, lets assume the underlying space $\mathcal{H}$ has finite dimension.

$$\mathcal{F}(\mathcal{H}) = \bigoplus_{k=0}^{\infty}\mathcal{H}^{\otimes k}$$

where by convention we have $\mathcal{H}^0 = \mathbb{C}\Omega$ for some unit vector $\Omega\in\mathcal{H}$

Remark 5. The induced inner product from $\mathcal{H}$ is defined by

$$(x,y)_{\mathcal{F}(\mathcal{H})} = (x_0\oplus x_1\oplus x_2\oplus ..., y_0\oplus y_1\oplus y_2\oplus ...)_{\mathcal{F}(\mathcal{H})} := \sum_{i=0}^{\infty}(x_i,y_i)_{\mathcal{H}^{\otimes i}}$$

Of course, in order to let $\mathcal{F}(\mathcal{H})$ well-defined, we require that the Fock space collects all finite norm, that is $\|x\|^2 = (x,x)_{\mathcal{F}(\mathcal{H})} < \infty$ and taking the completion of it.

Remark 6. To define the operator on $B(\mathcal{F}(\mathcal{H}))$. We require the Hilbert Schmidt norm to be bounded, that is, for $T \in B(\mathcal{F}(\mathcal{H}))$, we have $\|T\|_{HS} := \sup_{I\in\mathbb{N}} \sum_{i\in\mathbb{N}} (Te_i, e_i) < \infty$.

Definition 2. Given any vector $f \in \mathcal{H}$. The creation operator on Fock space is defined as $l(f) \in B(\mathcal{F}(\mathcal{H}))$ by

$$x = c\Omega \oplus x_1 \oplus x_2 \oplus ... \mapsto cf \oplus (f \otimes x_1) \oplus (f \otimes x_2) \oplus ...$$

Next given any linear operator on $T \in B(\mathcal{H})$. Define the guage operator $\Lambda(T) \in B(\mathcal{F}(\mathcal{H}))$ by

$$x = c\Omega \oplus x_{11} \oplus x_{21} \otimes x_{22} \oplus ... \mapsto 0 \oplus T(x_{11}) \oplus T(x_{21}) \otimes x_{22} \oplus ...$$

Last define the multiplication operator $\mu : \mathcal{H} \otimes \mathcal{H} \to \mathcal{H}$ by $x \otimes y \mapsto xy$

Remark 7. In the following text, the direct sum of elements will be denoted by a tuple of elements. That is,

$$x = c\Omega \oplus x_{11} \oplus x_{21} \otimes x_{22} \oplus ... = (c, x_{11}, x_{21} \otimes x_{22}, ...)$$

will be used if the context is not confusing.

Corollary 8. One can check that the adjoint operator of the creation operator $l(f)$ is $s(f)$ by

$$x = c\Omega \oplus x_{11} \oplus x_{21} \otimes x_{22} \oplus ... \mapsto (x_{11}, f)\Omega \oplus (x_{21}, f)x_{22} \oplus ...$$

In this case, we denote $l^*(f) = s(f)$

Proof. Let $x = (c, x_{11}, x_{21} \otimes x_{22}, ...)$ and $y = (c', y_{11}, y_{21} \otimes y_{22}, ...)$

$$\begin{aligned} (l(f)x, y) &= ((0, cf, f \otimes x_{11}, f \otimes x_{21} \otimes x_{22}, ...), (c', y_{11}, y_{21} \otimes y_{22}, ...)) \\ = (cf, y_{11}) + (f, y_{21})(x_{11}, y_{22}) + ... &= (c\Omega, (y_{11}, f)\Omega) + (x_{11}, (y_{21}, f)y_{22}) + ... \\ &= ((c, x_{11}, x_{21} \otimes x_{22}, ...), ((y_{11}, f), (y_{21}, f)y_{22}, ...)) = (x, s(f)y)) \end{aligned}$$

□

In our context, we note that we have $\mathcal{H}$ being an algebra. Next, define the multiplication operator $\mu : \mathcal{F}(\mathcal{H}) \to \bigoplus_{i=1}^{\infty} \mathcal{H}$ by

$$(c, x_{11}, x_{21} \otimes x_{22}, ...) \mapsto (0, x_{11}, x_{21}x_{22}, ..)$$

Tensor of modules

In the following context, I will study the M_n-modules. First, define the underlying space as $\mathcal{H} = M_n \otimes_{\mathbb{C}} M_m$. This, is the standard tensor product induced by bilinear form. Now, we will introudce the notation $\mathcal{H} \otimes_{M_n} \mathcal{H}$.

Definition 3. Given a ring R with unit 1_R. We say $(M, +)$ is a left R-module if $(M, +)$ is an abelian group with a mapping

$$\cdot : R \times M \to M$$

satisfying

1. $(r \cdot s) \cdot x = r \cdot (s \cdot x)$ for all $r, s \in R$ and $x \in M$
2. $r \cdot (x + y) = (r \cdot x) + (r \cdot y)$ for all $r \in R$ and $x, y \in M$
3. $(r + s) \cdot x = (r \cdot x) + (s \cdot x)$ for all $r, s \in R$ and $x \in M$
4. $1_R \cdot x = x$ for all $x \in M$

Remark 9. If the mapping $\cdot : M \times R \to M$ satisfying same condition, $(M, +)$ is called right R-module. Moreover, if $(M, +)$ is both left and right R-module, then we call $(M, +)$ a R-bimodule.

Corollary 10. In our setting, we only cosider for the case $\mathcal{H} = M_n(\mathbb{C}) \otimes M_m(\mathbb{C})$. Then, $\mathcal{H}$ is a M_n-bimodule with the unit $1_R = I_n$

Proof. Define the mapping $\cdot : M_n \times \mathcal{H} \to \mathcal{H}$ by

$$a \cdot (A \otimes B) \mapsto (aA) \otimes B$$

for $a \in M_n$ and $A \otimes B \in \mathcal{H}$. Similarly, we define another mapping $\cdot' : \mathcal{H} \times M_n \to \mathcal{H}$ by

$$(A \otimes B) \cdot a \mapsto (Aa \otimes B)$$

for $a \in M_n$ and $A \otimes B \in \mathcal{H}$. One can quickly verify they satisfy the definitions. $\square$

Remark 11. For a ring R and a left R-module M, we call M a free left R-module, if there is a basis of M with coefficient in R. For example, $\mathcal{H}$ is one of the case. Moreover, it is a free left and right M_n-module.

Proof. Given $x \in \mathcal{H}$. It has the for of

$$x = \sum_{k=1}^{h} c_k (N_k \otimes M_k) = \sum_{\alpha,\beta=1}^{m} \sum_{k=1}^{h} (c_k N_k) \cdot \left(I \otimes F_{\alpha\beta}\right) = \sum_{\alpha,\beta=1}^{m} \left(I \otimes F_{\alpha\beta}\right) \cdot \sum_{k=1}^{h} (c_k N_k)$$

It then follows that since the set $\left\{I \otimes F_{\alpha\beta}\right\}$ are linearly independent, $\mathcal{H}$ is free M_n -module. $\square$

Then, we are ready to construct our space of $\mathcal{H} \otimes_{M_n} \mathcal{H}$

Definition 4. For a free R-module, define the subspce $N \subset \mathcal{H} \otimes_{\mathbb{C}} \mathcal{H}$ by the span of

$$(x \cdot r) \otimes y - x \otimes (r \cdot y)$$

for all $x, y \in \mathcal{H}$ and $r \in M_n$. Now define $\mathcal{H} \otimes_{M_n} \mathcal{H}$ by

$$\mathcal{H} \otimes_{M_n} \mathcal{H} := \mathcal{H} \otimes \mathcal{H}/N$$

Last, define the left mapping $\cdot : M_n \times \mathcal{H} \otimes_{M_n} \mathcal{H} \to \mathcal{H} \otimes_{M_n} \mathcal{H}$ by

$$r \cdot (x \otimes y) \mapsto (r \cdot x) \otimes y$$

and right mapping $\cdot' : \mathcal{H} \otimes_{M_n} \mathcal{H} \times M_n \to \mathcal{H} \otimes_{M_n} \mathcal{H}$ by

$$(x \otimes y) \cdot r \mapsto x \otimes (y \cdot r)$$

Remark 12. $\mathcal{H} \otimes_{M_n} \mathcal{H} \cong M_n \otimes M_m \otimes M_m$ with isomorphism $\overline{\Phi} : \mathcal{H} \otimes_{M_n} \mathcal{H} \to M_n \otimes M_m \otimes M_m$ defined by

$$(A \otimes B) \otimes_{M_n} (C \otimes D) \mapsto AC \otimes B \otimes D$$

Proof. We will first check that it is well-defined. Let Φ be the mapping on $\mathcal{H} \otimes \mathcal{H}$ and π be the homomorphism from $\mathcal{H} \otimes \mathcal{H} \to \mathcal{H} \otimes_{M_n} \mathcal{H}$. Since it is easy to see this is actually a linear map on $\mathcal{H} \otimes \mathcal{H}$. Then, we only need to check $N = \ker \pi \subset \ker \Phi$. Then, $\Phi = \overline{\Phi} \circ \pi$ is well-defined.

By linearity of Φ, it sufficies to check for the case $n \in N$ of the form $n = (x \cdot r) \otimes y - x \otimes (r \cdot y)$ for some $r \in M_n$ and $x, y \in \mathcal{H}$ with x, y are of the seperable form. Then, $\Phi(n) = \Phi((x_1 r \otimes x_2) \otimes (y_1 \otimes y_2) - (x_1 \otimes x_2) \otimes (r y_1 \otimes y_2)) = 0$

Next we will show the map is bijective. For surjectivity, given any $X \otimes B \otimes D \in M_n \otimes M_m \otimes M_m$. We have $x = \pi((X \otimes B) \otimes (I_n \otimes D)) \in \mathcal{H} \otimes_{M_n} \mathcal{H}$ with $\overline{\Phi}(x) = \Phi((X \otimes B) \otimes (I_n \otimes D)) = X \otimes B \otimes D$. For injectivity, suppose $A_1 C_1 \otimes B_1 \otimes D_1 - A_2 C_2 \otimes B_2 \otimes D_2 = 0 \in M_n \otimes M_m \otimes M_m$. Then, by the freeness, we have $B_1 = B_2$ and $D_1 = D_2$ and $A_1 C_1 = A_2 C_2$. Next, $\pi((A_1 C_1 \otimes B_1) \otimes (I_n \otimes D_1) - (A_2 C_2 \otimes B_2) \otimes (I_n \otimes D_2)) = \pi((A_1 C_1 - A_2 C_2 \otimes B_1) \otimes (I_n \otimes D_1)) = 0$

Thus, we have checked that $\overline{\Phi}$ is an isomorphism. $\square$

Remark 13. $N \cong \ker \pi \cong \ker \mu \otimes M_m \otimes M_m \cong \mathcal{H} \otimes \mathfrak{sl}(n) \otimes M_m$

Proof. Since we can construct an homomorphism $\pi : \mathcal{H} \otimes \mathcal{H} \to \mathcal{H} \otimes_{M_n} \mathcal{H}$, we have $\ker \pi \cong N$. According to Remark 12, we have $\mathcal{H} \otimes_{M_n} \mathcal{H} \cong M_n \otimes M_m \otimes M_m$ and thus $\ker \pi \cong \ker \Phi$. Now, observe that $\Phi : \mathcal{H} \otimes \mathcal{H} \to M_n \otimes M_m \otimes M_m$ with $(A \otimes B) \otimes (C \otimes D) \mapsto AC \otimes B \otimes D$, then it is zero if and only if $AC = 0$. That is, $\ker \Phi \cong \ker \mu \otimes M_m \otimes M_m$

Recall that $\mu : M_n \otimes M_n \to M_n$ by $A \otimes B \mapsto AB$. Our goal here is to find $\ker \mu$. We claim that $\ker \mu = \operatorname{span}\{AB \otimes I_n - A \otimes B|\ A, B \in M_n\}$. The direction of $\operatorname{span}\{AB \otimes I_n - A \otimes B\} \subset \ker \mu$ is trivial. For the converse, lets suppose $P \otimes Q \in \ker \mu$. Then, $PQ = 0$. Since $P \otimes Q = \operatorname{tr}_n(Q) P \otimes I_n + P \otimes \mathring{Q} = (P(\operatorname{tr}_n(Q) I_n - Q)) \otimes I_n - P \otimes \mathring{Q} = P\mathring{Q} \otimes I_n - P \otimes \mathring{Q}$, it is of the desired form.

Note that $AB \otimes I_n - A \otimes B = A\mathring{B} \otimes I_n - A \otimes \mathring{B}$. We have $\operatorname{span}\{AB \otimes I_n - A \otimes B | A, B \in M_n\} = \operatorname{span}\left\{A\mathring{B} \otimes I_n - A \otimes \mathring{B}\right\} \cong M_n \otimes \mathfrak{sl}(n)$. The results then follows. □

Model

Now, we will use the same construction to build the Fock space over the ring M_n, denoted by

$$\mathcal{F}(\mathcal{H})_{M_n} := M_n \oplus \mathcal{H} \oplus \mathcal{H} \otimes_{M_n} \mathcal{H} \oplus \mathcal{H} \otimes_{M_n} \mathcal{H} \otimes_{M_n} \mathcal{H} \oplus ...$$

There exit a natural projection $\Pi : \mathcal{F}(\mathcal{H}) \to \mathcal{F}(\mathcal{H})_{M_n}$

Let us define two operations $O\left(C_\varphi\right), I\left(C_\varphi\right)$ on $B(\mathcal{F}(\mathcal{H}))$ by

$$O\left(C_\varphi\right)\xi : \mathbb{E}\left(C_\varphi\right)\xi$$

$$I\left(C_\varphi\right)\xi : C_\varphi \otimes \xi$$

Then, compute